ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · B4 · EARTH AND SPACE

Days, years and planets

★☆☆
Grade 5

Learning goals

- Recall facts about the Earth, Sun and Moon.
- Order the planets.

1 **Fill in the gaps.** [6]

WORD BANK

12 **24** **reflects** **axis** **west** **365¼** **east** **north**

The Earth spins on its axis once every 24 hours, causing day and night. It orbits the Sun once every 365¼ days. The Moon reflects light from the Sun. The Sun rises in the east and sets in the west.

2 **Number the planets in order from the Sun (1 = closest).** [4]

- 4 Mars
- 7 Uranus
- 1 Mercury
- 2 Venus
- 6 Saturn
- 3 Earth
- 5 Jupiter
- 8 Neptune

3 True or false? [5]

Statement	True	False	If false, correct it
1. The Sun moves around the Earth every day.		✔	The Earth spins; the Sun only appears to move.
2. The Moon makes its own light.		✔	It reflects sunlight.
3. When it is summer in Bangladesh it is winter in Australia.	✔		
4. Shadows are shortest at midday.	✔		
5. The Sun is a star.	✔		

4 Match each word to its meaning. [4]

Draw a line to match each pair, or write the letter.

rotation ●	→ A	● **A** one spin of the Earth on its axis
orbit ●	→ C	● **B** a year with 366 days
leap year ●	→ B	● **C** the path of the Earth around the Sun
constellation ●	→ D	● **D** a pattern of stars

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · B4 · EARTH AND SPACE

Explaining what we see

★★☆
Grade 5

Learning goals

- Explain day, night and seasons with diagrams.
- Use models.

1 **Shade the night side of the Earth and label day and night.** [2]

Sun on the left, Earth on the right

The half of the Earth facing away from the Sun is shaded and labelled night; the half facing the Sun is labelled day.

2 **Explain why it is daytime in Dhaka when it is night-time in New York.** [2]

They are on opposite sides of the Earth: when Dhaka faces the Sun, New York faces away from it.

3 **Explain why Bangladesh is hotter in June than in December. Use the words *tilted*, *direct* and *longer*.** [3]

In June the northern half of the Earth is tilted towards the Sun, so sunlight is more direct and the days are longer, giving more heat.

4 **A model: you shine a torch (Sun) at a ball (Earth) with a stick through it (axis) and turn the ball.** [3]

a) What does turning the ball show?

day and night (rotation)

b) Give **one** way this model is not like the real Earth and Sun.

Sizes and distances are wrong; the torch only lights one direction, but the Sun shines in all directions; the ball is not moving around the torch.

5 **Why does the Moon seem to change shape?** [2]

As it orbits the Earth we see different amounts of its sunlit half.

ANSWER KEY

INVESTIGATE · B4 · EARTH AND SPACE

How does day length change through the year?

★★☆ Grade 5

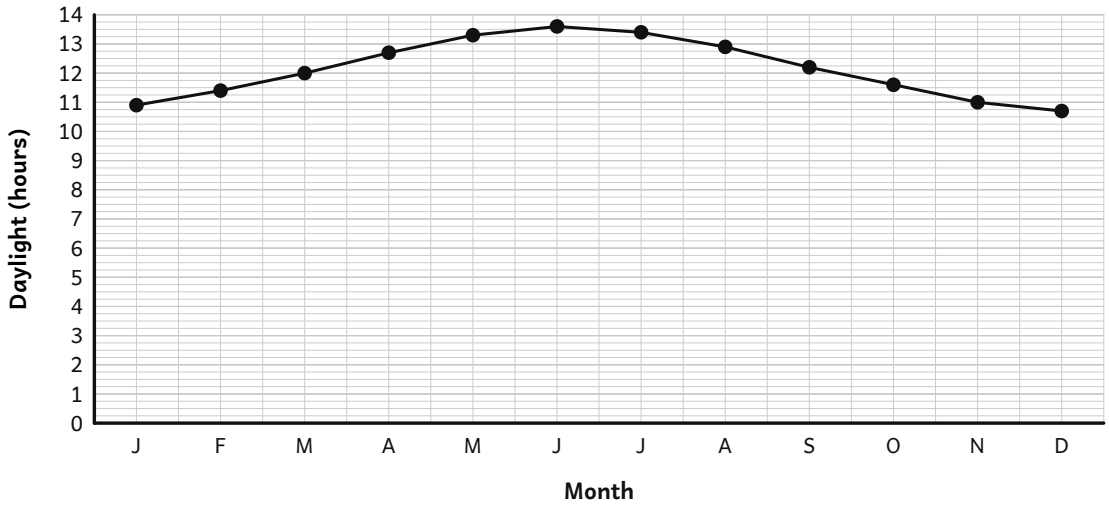
Learning goals

- Plot a line graph.
- Identify a pattern.
- Explain it using the Earth's tilt.

These are approximate hours of daylight in **Dhaka** in the middle of each month.

Month	J	F	M	A	M	J	J	A	S	O	N	D
Daylight (hours)	10.9	11.4	12.0	12.7	13.3	13.6	13.4	12.9	12.2	11.6	11.0	10.7

1 **Plot a line graph of daylight hours.** [4]



2 **Answer using your graph.** [4]

a) Which month has the longest days? the shortest?

longest: June; shortest: December

b) Describe the pattern.

Days get longer from December to June, then shorter again: a pattern that repeats every year.

c) How much longer is a day in June than in December?

13.6 – 10.7 = 2.9 hours (almost 3 hours)

3 **Predict the daylight hours in Singapore (on the equator) in June and in December. Explain.** [2]

About 12 hours in both months: near the equator day length hardly changes during the year.

4 **Shadow stick. Plan how you would measure the shadow of a stick at different times of day. What would you keep the same?** [2]

Same stick, same place, measure at 9 am, 11 am, 1 pm, 3 pm with a metre stick; record length and direction; repeat on another sunny day.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · B4 · EARTH AND SPACE

Six seasons, one planet

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Grade 5

Learning goals

- Compare seasons in different places.
- Link seasons to culture and farming.
- Explain using the Earth's tilt and position.

Science · Geography · Culture · Maths · Agriculture

Bengali tradition divides the year into **six seasons** (*ritu*): *Grishmo* (summer), *Borsha* (monsoon), *Shorot* (early autumn), *Hemonto* (late autumn), *Sheet* (winter) and *Boshonto* (spring). Farming follows the seasons: *Aman* rice grows in the monsoon and is harvested in Hemonto, celebrated with the **Nabanna** festival of new rice, while *Boro* rice grows in the dry winter with water from tube-wells.

Other places have very different seasons. In **Singapore**, near the equator, it is hot all year and every day has about 12 hours of daylight. In **Tromsø**, Norway, far to the north, the Sun does not set for about two months in summer and does not rise for about two months in winter. In **Sydney**, Australia, south of the equator, December is the middle of summer.

Day-length facts from standard astronomical tables (approximate).

1 **Complete the table.** [6]

Place	Where it is	What its seasons are like
Dhaka	north of the equator, near the Tropic of Cancer	hot, wet summer (monsoon); mild, dry winter; days 11–13.5 h
Singapore	on the equator	hot all year; about 12 h daylight every day
Tromsø	far north, near the Arctic	midnight sun in summer, polar night in winter
Sydney	south of the equator	summer in December, winter in June

2 **Explain why Sydney has summer in December, when Dhaka has winter.** [2]

In December the southern half of the Earth is tilted towards the Sun (summer there), while the northern half, with Dhaka, is tilted away (winter).

3 **Choose one Bangladeshi festival or food linked to a season. Explain the link.** [2]

e.g. Nabanna: celebrates the new Aman rice harvest in Hemonto; pitha in winter made with new rice and date-palm molasses; mangoes in Grishmo.

4 **Why does Boro rice need water from tube-wells, but Aman rice does not?** [2]

Boro grows in the dry winter with little rain; Aman grows in the monsoon when rain supplies the water.

ANSWER KEY

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REVIEW AND CHALLENGE · B4 · EARTH AND SPACE

Show what you know

★★★
Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

R	S	U	N	W	P	Y	G	L	J
T	T	A	P	R	M	R	K	G	I
E	M	M	I	N	W	S	T	E	T
Q	Y	B	O	R	B	I	T	R	B
U	F	Y	C	O	H	I	B	W	J
A	R	G	F	W	N	O	F	C	H
T	I	L	T	T	M	S	T	A	R
O	J	L	P	L	A	N	E	T	Y
R	G	H	S	E	A	S	O	N	H
K	L	A	X	I	S	R	D	J	M

axis **equator** **moon** **orbit** **planet** **season** **star** **sun** **tilt**

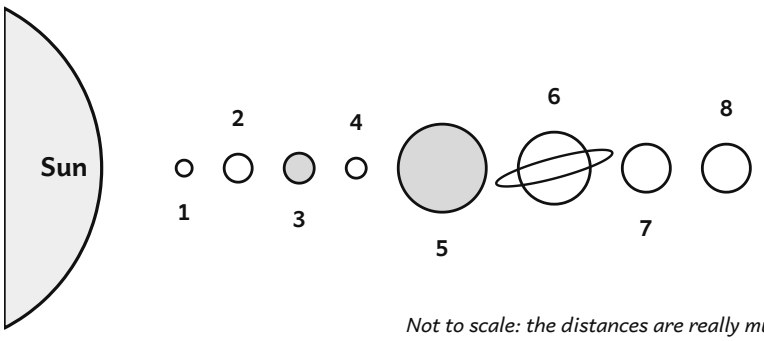
2 **What causes day and night?** [1]

- ○ the Earth orbiting the Sun
- ● **the Earth spinning on its axis**
- ○ the Moon blocking the Sun
- ○ clouds

3 **What causes the seasons?** [1]

- ○ the Earth's distance from the Moon
- ● **the tilt of the Earth's axis as it orbits the Sun**
- ○ the Sun getting hotter
- ○ the wind

4 **Label the planets with numbers 1 to 8 from the Sun, then write the name of the largest planet.** [2]



Not to scale: the distances are really much, much bigger.

Jupiter

5 **Challenge. Galileo saw moons going around Jupiter. Why was this important evidence?** [2]

It showed that not everything orbits the Earth, which supported the idea that the Earth and planets orbit the Sun.